

Optimizing ad-placement for socio-demographic segments using mobility data

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1. INTRODUCTION

Optimizing ad placement is a central challenge at the intersection of data science, urban mobility, and personalized marketing. In this work we propose a predictive model that, using anonymized mobility data collected in the Île-de-France region (Paris and surroundings), estimates the probability that individuals belonging to specific socio-demographic segments will go through an urban area. Our aim is to generate probabilistic heatmaps that identify, for a socio-demographic segment, the areas most commonly visited, thus enabling strategic decision-making in geolocated ad-placement campaigns.

2. METHODOLOGY

A. Dataset Description

This study uses data from the 2023 EMG Survey, which includes one week of GPS tracking for 3,337 Île-de-France residents aged 16 to 80. The dataset comprises socio-demographic information, trip details, and raw GNSS traces. In this preliminary stage, we focus on the individual data and GNSS traces, limiting the analysis to GPS points within Paris.

B. Target Construction

The primary objective of this study is to estimate the probability that individuals will visit different areas of the city. As this information is not directly available in the dataset, we developed a pipeline to infer it from the GNSS data.

In this preliminary stage, we focused on estimating visit probabilities for each individual in the dataset. For every user, we collected all GNSS traces within the city of Paris, and we fitted a Gaussian Mixture Model (GMM) to capture the spatial distribution of their movements. The optimal number of clusters was determined using the Bayesian Information Criterion (BIC), with a mean of 4.5 clusters across all users.

Subsequently, the city was divided into a 50x50 spatial grid and, for each user, we computed the probability of visiting each grid cell based on their personal GMM distribution. These probability values were then used as the target variable in the predictive modeling stage.

C. Feature Engineering

To capture meaningful subgroups within the population and uncover latent behavioral profiles, Non-negative Matrix Factorization (NMF) was applied to the demographic data. This decomposition revealed hidden structures and enriched the feature space, resulting in 10 additional features included in the model.

In addition, minor preprocessing steps were applied to ensure data consistency. Missing values were handled by assigning default values per column, and categorical variables were label-encoded to ensure all features were represented as floats.

D. Dataset Split and Data Leakage Prevention

The dataset was split into 80% for training and 20% for testing. To prevent data leakage, all observations corresponding to a given user were assigned exclusively to either the training or the test set, ensuring no overlap between user data across splits.

E. Predictive Modeling and Training

In this preliminary stage, the model selected to predict the target variable was XGBoost, a gradient-boosted tree-based algorithm. This choice was motivated by XGBoost's strong performance on tabular data, its ability to capture complex non-linear relationships, and its built-in feature importance capabilities. Given that the target is defined over spatial grid blocks derived from latitude and longitude, we considered XGBoost well-suited to distinguish between these discrete spatial regions. Root Mean Squared Error (RMSE) was used as the loss function during training.

A hyperparameter search was conducted using K-Fold cross-validation on the training set, carefully ensuring that no user data leaked across folds.

Finally, the model was retrained on the full training set using the best hyperparameters, and its performance was evaluated on the held-out test set.

F. Prediction Validation

To evaluate model performance, the predicted heatmap for each user was compared against their ground-truth mobility distribution, which was represented by the Gaussian Mixture Model (GMM) fitted to their observed GNSS traces. This comparison served both as a benchmark to track improvements during training and as the basis for computing the model's loss function.

It is important to clarify, however, that our goal is not to replicate a user's exact mobility pattern. A model that perfectly reproduces an individual's GMM would risk severe overfitting, merely memorizing their specific trajectory during the measurement period. Instead, our objective is to generalize across users with similar socio-demographic characteristics. We aim for the predicted heatmap to cover the regions visited by the user, without exactly replicating their personal route. In doing so, the model captures broader behavioral tendencies of the target group rather than the unique patterns of any single individual, which is essential for scalable and privacy-aware applications such as targeted advertising. If a user's predicted heatmap was exactly the same as their GNSS trace, it would be at the cost of excluding users with similar characteristics who took different routes.

3. CASE STUDIES

To illustrate the practical value of our model, we present two case studies that simulate real-world advertising use cases. In both cases, we use the model to estimate the spatial probability distribution of mobility for individuals belonging to specific

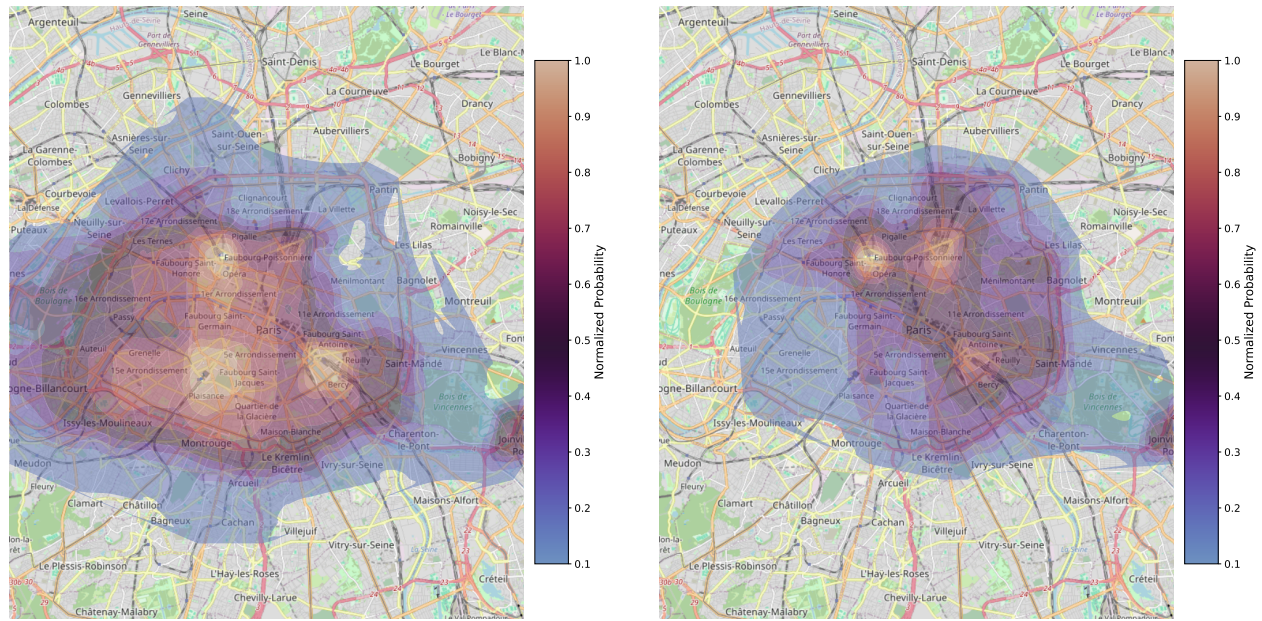


Fig. 1. Predicted distributions for two socio-demographic profiles: (Left) BMW buyers; (Right) Households with persons over 65.

socio-demographic segments. The first case considers users with demographic profiles consistent with high-end automotive consumers, specifically, potential BMW buyers. The second profile targets individuals living in households with older adults, relevant for advertisement campaigns related to elder care services or products. These examples demonstrate how probabilistic mobility predictions can guide spatial targeting in ad-placement strategies, identifying zones with a higher likelihood of exposure for each consumer type.

To construct the BMW buyer profile, we drew on demographic insights reported by Hedges and Company [2], which characterizes BMW owners as typically male, aged 50 or older, and employed in executive or senior intellectual professions. Based on these criteria, we identified a subset of users in our test set who met these conditions. For each selected individual, the model predicted a probability distribution over spatial centroids. These predictions were then weighted using the individual's survey weight (as provided in the original dataset), and the resulting values were normalized to produce a population-level probability map. This map reflects the aggregated likelihood of individuals matching the BMW buyer profile being present in each spatial region. While this approach is constrained by the availability of matching individuals within the test set, future work will address this limitation by shifting from test-set filtering to direct prediction over user clusters defined by socio-demographic conditions. This would enable ad planners to generate probabilistic mobility maps for custom target audiences, regardless of their representation in the original dataset.

As a second case study, we modeled a socio-demographic profile representing individuals living in households with elderly members, specifically those aged 65 or older. To identify relevant users in the test set, we applied the following filtering criteria: individuals aged 30 or older, living with at least one person aged 65 or above, and classified under the “worker” professional category. These constraints are illustrative and serve as a proxy for potential consumers of eldercare products or services; in practice, advertising agencies may define such profiles differently depending on their campaign objectives. For each qualifying user, the same procedure described in the previous case was applied to generate a composite mobility map for the

target group.

As shown in Figure 1, the predicted mobility distributions differ substantially between the two socio-demographic profiles. The profile associated with potential BMW buyers exhibits a more spatially dispersed pattern, indicating broader movement across the city. This suggests that individuals in this segment may have greater access to private transportation and fewer mobility constraints. The highest-probability areas in the map are close to Boulevard Haussmann and Faubourg Saint-Honoré. In contrast, the mobility of individuals living in households with a person over 65 is more spatially concentrated, likely reflecting limitations related to transportation mode or physical ability. In this case, the areas with highest-probability are Montparnasse and the area close to Gare de Lyon. These distinct patterns highlight the model's capacity to capture relevant behavioral differences across segments. From a marketing perspective, such insights can inform targeted ad-placement strategies by identifying high-affinity zones and eliminating low-yield areas where the target audience is unlikely to be present.

4. FUTURE WORK

This preliminary study covered the City of Paris; in our full report we plan to extend the analysis to the full Île-de-France region. Our machine learning model was trained to make user-level predictions of the spatial distribution of the trajectories, by leveraging socio-demographic information. Users are clustered using non-negative matrix factorization (NMF). As part of the next steps, we will consider averaging similar users as input for the model in order to avoid overfitting. We will also explore alternative learning structures to directly predict the aggregated flows from these clusters, e.g. by extending the deep gravity model in [1] to incorporate these profiles as input.

Disclosures. The authors declare no conflicts of interest.

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